

PENUMBRA

Quality Assurance Report — Release Readiness

Document ID: PEN-QA-001 **Version:** 1.0 **Date:** 24 July 2026

Verdict: READY FOR TESTING & DEPLOY

Baseline: PEN-SRS-001 v1.0

1. Summary

PENUMBRA v1.0 was verified against the SRS acceptance criteria. The automated suite (**35 tests, 35 passed, 0 failed**, ~0.8 s) covers the F10.7 forecast, Kp categorical forecast, the leakage-free calibration backtest, the drag translation, and the full API contract. The UI builds clean under strict TypeScript and renders the fan chart, Kp strip, drag panel, and calibration diagrams correctly. Live NOAA connectivity was confirmed. Two findings surfaced during QA (both resolved). **Verdict: ready for testing and deployment alongside KESSLER.**

2. Automated Test Results

Suite	Coverage	Result
test_f107_forecast (8)	Point reverts toward climatology at long lead; band monotonicity $p_{05} \leq \dots \leq p_{95}$ per lead; 64-sfu floor; synthetic flat-signal recovery; bands widen with lead.	PASS
test_kp_forecast (6)	Category boundaries; probabilities sum to 1 per lead; recent-storm sensitivity; Ap band ordering; full-forecast shape.	PASS
test_calibration (9)	Leakage-freedom (corrupting the future cannot change past-origin errors); nominal coverage on stationary synthetic; pinball zero at perfect quantile and correct asymmetry; skill sign; reliability on a perfectly-calibrated synthetic.	PASS
test_drag (5)	Density increases with F10.7; altitude ordering (lower = denser); reference value at F_REF; higher-flux widens decay band; lower altitude → more decay.	PASS
test_api (7)	Health; F10.7 contract + band monotonicity + floor + NOAA overlay; Kp probability normalisation; drag altitude ordering; calibration shape; observations; briefing fallback; pipeline status; validation + 404 envelopes.	PASS

3. Live Verification

Check	Observed	Status
Backend boot (Demo Mode, zero credentials)	Pipeline ran on startup; all 6 agents report <code>ok</code> ; forecast/calibration snapshots published.	✓
GET /forecast/f107	45-day point + p05–p95 bands, monotone, floored; NOAA overlay present.	✓
GET /forecast/kp	7-day category probabilities summing to 1; expected Ap band.	✓
GET /forecast/drag	±1904 km in-track at 400 km, ±248 km at 550 km, ±12 km at 800 km — correct altitude dependence.	✓
GET /calibration	90% empirical coverage (target 0.90); beats persistence at 36/45 leads; honest skill dips shown.	✓
Live NOAA connectivity	Real fetch returned 122 F10.7 rows (last 128 sfu) and 62 Kp rows — live mode functional.	✓
UI (strict TS build + render)	Fan chart, Kp strip, drag panel, calibration diagrams render; 60 KB gzipped, no chart library.	✓
Briefing graceful degradation	No API key → deterministic template briefing, <code>source: "template"</code> , no error.	✓
Error envelope + observability	Validation → 422 envelope; unknown API route → 404 envelope; request-id in logs and X-Request-Id header; rotating file log active.	✓

4. Findings

ID	Severity	Finding	Disposition
QA-1	Major	SPA catch-all masked API 404s. With <code>ui/dist</code> present, the single-service SPA fallback served <code>index.html</code> (HTTP 200) for unmatched <code>/api/*</code> routes instead of 404.	Fixed: the SPA handler now returns the 404 error envelope for <code>api/</code> , <code>docs</code> , <code>openapi.json</code> , <code>redoc</code> prefixes. Regression covered by <code>test_unknown_route_404</code> .
QA-2	Note	Skill measured vs persistence, not NOAA historical. NOAA's free endpoint exposes only the current 45-day forecast, not a historical archive, so a reproducible backtest cannot score against past NOAA issues.	By design: skill is scored against a reproducible persistence baseline (clearly labelled), and NOAA's live forecast is overlaid on the forecast chart for direct current-issue comparison. Documented in the calibration summary and HLD §3.
QA-3	Note	Honest skill dips on demo data. On the strongly-periodic bundled demo series, mean-reversion underperforms persistence at a few leads (visible negative skill).	Retained deliberately — showing where the model is weaker is the credibility point; real F10.7 gives a smoother profile.
QA-4	Note	Drag translation is a reduced exponential-atmosphere model.	Explicitly labelled "reduced-model, order-of-magnitude" in the UI and API (FR-DRAG-3).

5. Requirements Traceability (Must-haves)

FR-ING-1..6 ✓ (demo fallback, TTL cache, last-known-good; live NOAA verified) · FR-F107-1..4 ✓ (point + empirical-quantile bands, monotone, floored) · FR-KP-1..3 ✓ · FR-VER-1..5 ✓ (leakage-free backtest, coverage, reliability, pinball, skill, all exposed in UI) · FR-DRAG-1..3 ✓ · FR-API-1..3 ✓ · FR-UI-1..6 ✓ (fan chart, Kp strip, drag, calibration page, pipeline, licence in UI) · NFR-1/2/4/7 ✓ · NFR-3 spot-checked (backtest < 1 s on bundled record) · NFR-5 (coexistence) to be confirmed on deploy · NFR-6 (logic-layer coverage) — comprehensive suites in place.

6. Sign-off

QA verdict: **APPROVED for testing and deployment.** Deployment must run PENUMBRA as an isolated systemd service on its own port with the co-hosted KESSLER service unaffected (NFR-5), to be verified during the deploy step.

